

Reconfigurable Microwave Photonic Demultiplexer Using Integrated CROW and Lattice Filters

Luis Gonzalez-Guerrero⁽¹⁾, Alberto Zarzuelo⁽¹⁾, Jiale Tian⁽¹⁾, Jose Manuel Delgado Mendinueta⁽¹⁾, Charoula Mitsolidou⁽²⁾, Chris G.H. Roeloffzen⁽²⁾, Guillermo Carpintero⁽¹⁾

⁽¹⁾ Universidad Carlos III de Madrid, Departamento de Tecnología Electrónica, Avenida de la Universidad 30, 28911 Leganes, Madrid, Spain, lgguerre@ing.uc3m.es

⁽²⁾ LioniX International BV, Hengelosestraat 500, 7521 AN Enschede, The Netherlands

Abstract We demonstrate an integrated microwave photonic demultiplexer using cascaded CROW and lattice filters, enabling tunable bandwidth (125–500 MHz) and strong channel suppression. Experimental demultiplexing of 16-QAM signals at 23 GHz is demonstrated, confirming reconfigurable filtering for flexible satellite payloads. ©2026 The Author(s)

Introduction

The increasing demand for flexible satellite payloads is driving the development of RF channelization solutions capable of dynamically selecting and routing spectral resources across the Ka, Q, and V bands [1, 2]. However, conventional RF implementations based on filter banks and reconfigurable filtering modules are typically constrained to predefined frequency bands and offer limited tunability, while facing challenges in meeting the stringent size, weight, and power (SWaP) requirements of satellite payloads and scaling efficiently to higher frequencies [3, 4].

Microwave photonics has emerged as a promising solution for broadband RF signal processing by exploiting the intrinsic advantages of optical systems, including wide bandwidth, low transmission loss, and immunity to electromagnetic interference. In particular, photonic payload architectures have shown strong potential for flexible telecom satellites and next-generation 5G satellite networks [5, 6]. However, the implementation and experimental assessment of photonic payloads remain limited. In this context, integrated microwave photonics plays a key role by enabling compact, scalable, and reconfigurable filtering functionalities suitable for practical implementations. Building on this, this work experimentally demonstrates a microwave photonic demultiplexer for RF channel selection with reconfigurable filter bandwidth, with performance validated through the filtering of 16-QAM signals of different symbol rates.

Experimental arrangement

Fig. 1(a) illustrates the demultiplexer architecture, which consists of an external cavity laser (ECL), a phase modulator, a photodiode (PD), and two silicon nitride circuits: an eight-coupled ring resonator optical waveguide (CROW) filter followed by a six-stage asymmetric Mach–Zehnder interferometer (AMZI) lattice filter, both fabricated by

LioniX International.

To understand the role of each filter in the demultiplexer, the operation of the system is described next. The ECL generates an optical carrier that is split into two paths by a 50/50 coupler. In the first path, the light is directed to a phase modulator, where an incoming RF signal containing both the target and two adjacent dummy channels is imposed onto the optical carrier, generating a double-sideband (DSB) signal with optical carrier. This modulated signal is then passed through the CROW filter, which selects the target channel from the upper modulation sideband. However, due to the limited free spectral range (FSR) of the CROW filter, part of the unwanted lower sideband, as well as the optical carrier, may also be transmitted.

To address this limitation, the output of the CROW filter is fed into the lattice filter. This filter performs two functions: (i) it suppresses the unwanted lower sideband and the residual optical carrier, and (ii) it reinjects the optical carrier coming from the second path of the 50/50 coupler after the ECL. The output of the lattice filter, which contains only the target channel and the optical carrier, is sent to the PD, which converts the signal back to the RF domain, as shown in inset (4) of Fig. 1(a).

Filter characterization

The lattice filter consists of a six-stage AMZI structure with tunable couplers and phase shifters, enabling a reconfigurable filter order. In this work, it is operated as a fifth-order filter. A detailed description and characterization are provided in [7]. Fig. 1(c) depicts the packaged module containing the lattice filter, while Fig. 2(a) illustrates its spectral response when configured as a fifth-order filter.

The CROW filter consists of eight coupled ring resonators with tunable couplers and phase shifters, which enable central frequency tuning over its 2.48-GHz FSR, reconfigurable bandwidth

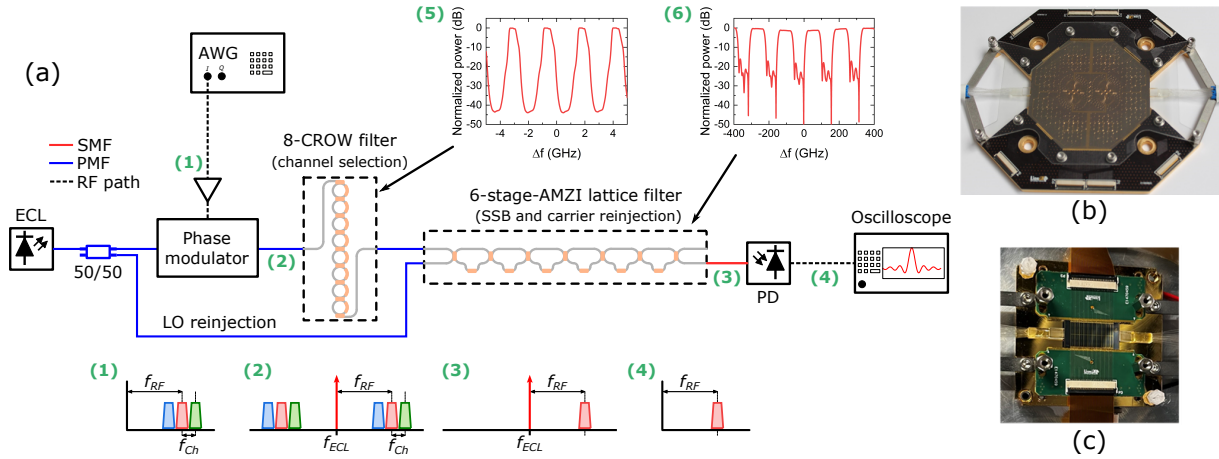


Fig. 1: (a) Microwave photonic demultiplexer comprising an external cavity laser (ECL), a phase modulator, a coupled ring resonator optical waveguide (CROW) filter, an asymmetric Mach–Zehnder interferometer (AMZI) lattice filter, and a photodiode (PD). Insets (1)–(4) show the signal spectrum at different points of the demultiplexer, while insets (5) and (6) show the normalized response of the CROW and AMZI filters, respectively. (b) Packaged module containing the CROW filter. (c) Packaged module containing the AMZI filter. AWG: arbitrary waveform generator, SMF: single mode fiber, PMF: polarization-maintaining fiber, f_{Ch} : frequency offset between channels, f_{RF} : carrier frequency of the target channel.

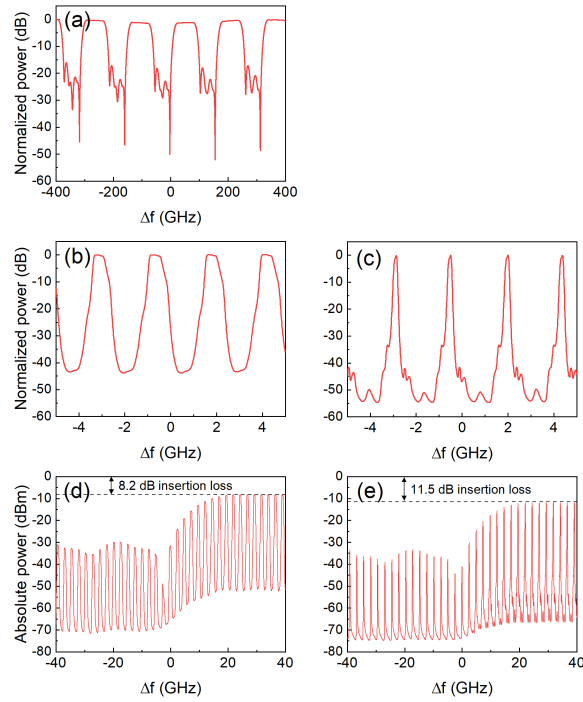


Fig. 2: Normalized frequency response of (a) the lattice filter and (b) and (c) the CROW filter for passband bandwidths of 500 MHz and 125 MHz, respectively. Absolute frequency response of the lattice and CROW filters connected in series for CROW passband bandwidths of (d) 500 MHz and (e) 125 MHz. All frequency offsets are measured with respect to the frequency associated to a wavelength of 1554.8 nm

tuning from 42 to 1500 MHz, and an 8th-order spectral response. Fig. 1(b) shows the packaged module containing the CROW filter.

Fig. 2(b) and (c) show the normalized frequency response of the CROW filter for passband bandwidths of 500 MHz and 125 MHz, respectively. A rejection ratio of approximately 40 dB is achieved for the 500 MHz configuration, while a higher rejection of 55 dB is obtained for the 125 MHz configuration. The presence of multiple spurious peaks in Fig. 2(c) suggests a partial misalignment of the

CROW rings, resulting in an effective filter order lower than the ideal eighth order; however, this response can be further optimized through fine tuning.

Fig. 2(d) and (e) show the corresponding absolute frequency responses of the lattice and CROW filters connected in series. As shown, the total insertion loss, including both filters, was measured to be 8.2 dB and 11.5 dB for the 500 MHz and 125 MHz CROW filter bandwidth configurations, respectively. In addition to the filter losses, an extra 5.58 dB of loss was measured in the signal path, while the LO reinjection path exhibited a loss of 6.7 dB. The laser output power was 12 dBm, and the responsivity of the photodiode was 0.65 A/W.

Transmission experiments

For the transmission experiments, the setup shown in Fig. 1(a) was used. Two experiments were performed: one with the CROW filter set to a 500 MHz bandwidth and another with the CROW filter set to a 125 MHz bandwidth. The lattice filter configuration remained unchanged during the two experiments (i.e., as shown in Fig. 2(a)). In both experiments, a total of three channels were generated with the arbitrary waveform generator (AWG): one target channel to be demultiplexed at the receiver and two adjacent dummy channels to emulate neighboring channel interference. The carrier frequency of the target channel (f_{RF}) was approximately 23 GHz in both cases, limited by the bandwidth of the RF amplifier after the AWG. The frequency offset between adjacent channels (f_{Ch}) varied from 1.25 GHz for the 500 MHz bandwidth configuration to 0.8 GHz for the 125 MHz configuration.

All channels, including the dummy channels, used 16-quadrature amplitude modulation (16-

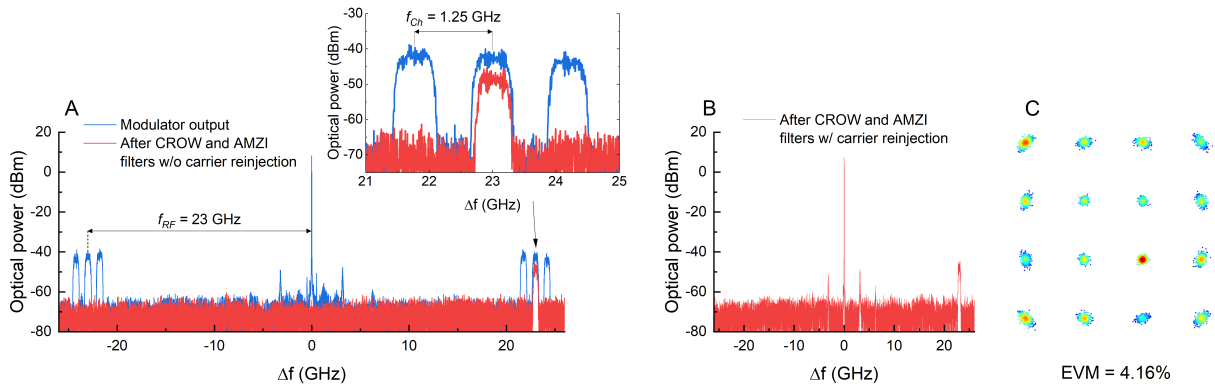


Fig. 3: Transmission spectra obtained for a CROW filter passband bandwidth of 500 MHz: (a) after the phase modulator (blue) and after the CROW and lattice filters without carrier reinjection (red); (b) after carrier reinjection; (c) corresponding constellation diagram and error vector magnitude (EVM) after digital demodulation.

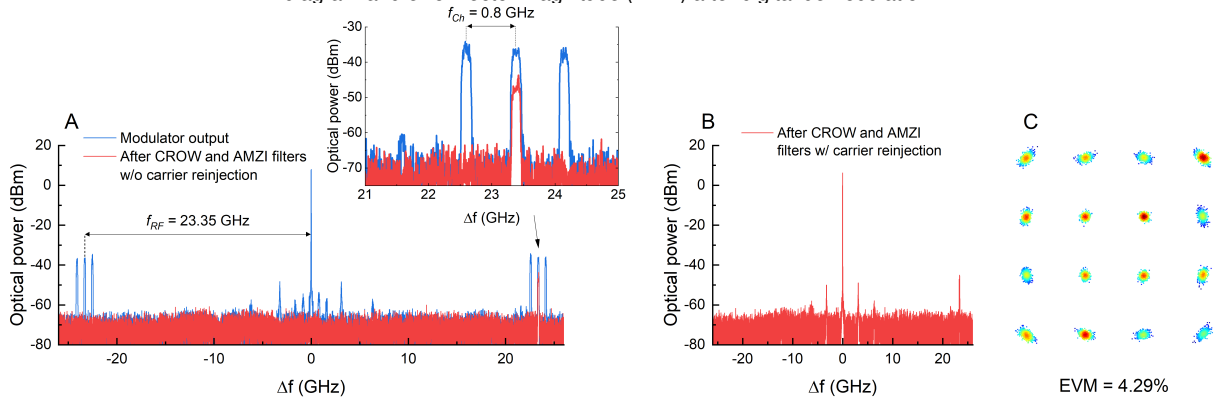


Fig. 4: Transmission spectra obtained for a CROW filter passband bandwidth of 125 MHz: (a) after the phase modulator (blue) and after the CROW and lattice filters without carrier reinjection (red); (b) after carrier reinjection; (c) corresponding constellation diagram and error vector magnitude (EVM) after digital demodulation.

QAM) with symbol rates of 500 MBd and 125 MBd for the two CROW filter bandwidth configurations. A root-raised-cosine (RRC) filter with a roll-off factor of 0.35 was applied in both cases, resulting in signal bandwidths of 675 MHz and 165.75 MHz, respectively. Although these bandwidths exceed the CROW filter passband bandwidths, a low error vector magnitude (EVM) was still achieved, as discussed next. A total of 10^4 symbols were used for EVM analysis.

Fig. 3(a) shows the spectrum after the phase modulator (blue) and after the CROW and lattice filters (red) for a CROW filter passband bandwidth of 500 MHz and when no carrier reinjection is performed. The combination of the two filters effectively suppresses the adjacent dummy channels, as well as the carrier and the lower sideband. The inset of Fig. 3(a) shows that the filter preserves the original signal shape, with attenuation only due to the insertion loss of the filters, measured to be 8.2 dB (see Fig. 2(d)).

Fig. 3(b) shows the output of the filters when carrier reinjection is performed. The carrier is attenuated by 2.5 dB after reinjection, highlighting the advantage of this approach compared to using a 3 dB coupler, where both the carrier and the signal would experience a 3 dB loss. After digitization and basic digital signal processing, an error

vector magnitude (EVM) of 4.16% was obtained, as shown in Fig. 3(b). Residual phase noise is observed, likely due to partial decorrelation of the laser tones travelling through the signal and LO reinjection paths, as well as the increased sensitivity of low symbol rates to phase noise. This impairment could be mitigated by employing more advanced phase noise compensation algorithms, such as the blind phase search [8], instead of the modified Viterbi–Viterbi algorithm [9] used in this work.

Fig. 4(a) shows the spectrum after the phase modulator (blue) and after the two filters without carrier reinjection (red) for the 125 MHz transmission experiment, while Fig. 4(b) shows the corresponding spectrum with carrier reinjection for the same bandwidth. In this case, an EVM of 4.29% was obtained as shown in Fig. 4(c).

Conclusions

An integrated microwave photonic demultiplexer with tunable bandwidth and strong channel selectivity is experimentally demonstrated. Low-EVM demultiplexing of 16-QAM signals at 23 GHz confirms its potential for flexible and reconfigurable satellite payloads.

Acknowledgements

This work was supported by the European Space Agency (ESA) through the TRP project THOR-MUX, by the European Union's Horizon Europe research and innovation programme through the Smart Networks and Services Joint Undertaking (SNS JU) under Grant Agreement No. 101096949 (TERA6G project), and by the Spanish Ministry of Economic Affairs and Digital Transformation through the PERTE Chip initiative (EPICPack Chair, Ref. TSI-069100-2023-6).

References

- [1] O. Kodheli et al., "Satellite communications in the new space era: A survey and future challenges", *IEEE Communications Surveys and Tutorials*, vol. 23, no. 1, pp. 70–109, 2021. DOI: 10.1109/COMST.2020.3028247.
- [2] S. Pan et al., "Satellite payloads pay off", *IEEE Microwave Magazine*, vol. 16, no. 8, pp. 61–73, 2015. DOI: 10.1109/MMM.2015.2441619.
- [3] M. Li, Z. Wang, J. Liu, and Y. Luo, "Switching filter bank module based on advanced rf system-in-package", in *International Conference on Microwave and Millimeter Wave Technology*, Beijing, China, 2024, pp. 1–3. DOI: 10.1109/ICMMT61774.2024.10672376.
- [4] W. Chen, L.-F. Qiu, and L.-S. Wu, "Reconfigurable filter with switchable operating frequency, bandwidth, and response type using rf switches and circulators", *IEEE Transactions on Microwave Theory and Techniques*, vol. 73, no. 3, pp. 1708–1716, 2025. DOI: 10.1109/TMTT.2024.3448497.
- [5] M. Sotom, M. Aveline, R. Barbaste, B. Benazet, A. L. Kernec, et al., "Flexible photonic payload for broadband telecom satellites: From concepts to system demonstrators", in *Proceedings of SPIE*, vol. 10562, 2017. DOI: 10.1117/12.2296157.
- [6] V. Polo, M. A. Piqueras, and J. Martí. "Photonic payloads for next-generation 5g satellite networks", *IEEE Future Networks*. [Online]. Available: <https://futurenetworks.ieee.org/tech-focus/november-2019/photonic-payloads-for-next-generation-5g-satellite-networks>.
- [7] C. Mitsolidou et al., "Optical single-sideband (ssb) lattice filter enabling a microwave photonic demultiplexer", in *Proceedings of SPIE*, vol. 13699, 2025. DOI: 10.1117/12.3075202. [Online]. Available: <https://doi.org/10.1117/12.3075202>.
- [8] T. Pfau, S. Hoffmann, and R. Noe, "Hardware-efficient coherent digital receiver concept with feedforward carrier recovery for m-qam constellations", *J. Lightwave Technol.*, vol. 27, no. 8, pp. 989–999, 2009.
- [9] M. Seimetz, "Laser linewidth limitations for optical systems with high-order modulation employing feed-forward digital carrier phase estimation", in *OFC*, Paper OTuM2, 2008.